

# Enhancing Trusted Multi-View Classification via Adaptive Regularization Guided by View-Specific Biases

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## Abstract

Trusted multi-view classification (TMC) aims to improve prediction reliability by integrating evidence from multiple views. Existing TMC methods extract evidence from single view and use a regularization term to shape the evidence distribution. However, existing methods typically enforce a uniform regularization objective across all views, overlooking critical view-specific biases: intra-view class ambiguity caused by confusable features and inter-view quality disparities reflected in evidence uncertainty. To address these issues, we propose an adaptive regularization strategy that enhances robustness on two levels. At the intra-view level, it quantifies feature ambiguity to apply targeted relaxation to confusable classes, preventing over-penalization of inherent uncertainty. At the inter-view level, it evaluates relative view quality to impose stronger constraints on unreliable views and suppress noise from low-quality ones. Extensive experiments across multiple benchmarks demonstrate the superiority and reliability of the proposed method.

## CCS Concepts

• Computing methodologies → Supervised learning by classification; Neural networks.

## Keywords

Trusted Multi-View Classification, Adaptive Regularization, View-Specific Biases

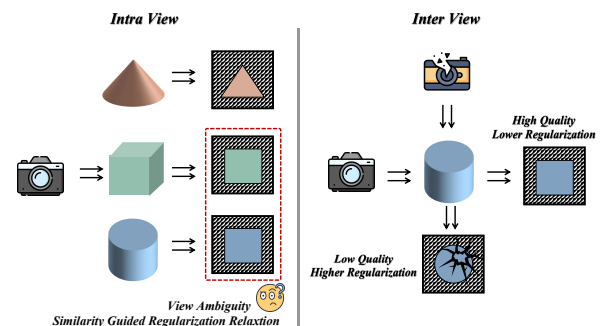
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**Figure 1: Multi-view challenges: (Left) intra-view ambiguity where certain views induce class confusion; (Right) inter-view quality disparity where views vary in reliability.**

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## 1 Introduction

Multi-view perception is central to **risk-sensitive applications** like autonomous driving and medical imaging [3, 11, 18], where systems fuse complementary cues (e.g., RGB, depth, LiDAR) for decision making. In these scenarios, communicating *why* a system is uncertain—due to noisy views, missing data, or intrinsic class ambiguity [21]—is as critical as predictive accuracy.

Evidential deep learning (EDL) within the Trusted Multi-View Classification (TMC) framework is a primary tool for this, as it allows networks to output "evidence" parameterizing a Dirichlet distribution. The training objective in TMC is typically twofold: an evidential classification loss serves to maximize evidence for the correct (target) class, while a Kullback–Leibler (KL) regularizer is used to constrain and penalize the evidence assigned to non-target classes. This KL term is central to the framework's goal of quantifying uncertainty, and as prior work confirms, its specific design is a critical factor in model robustness and calibration [2].

However, representative TMC methods universally adopt a uniform KL design, which introduces two fundamental limitations, as shown in Figure 1:

- (1) **Intra-View Ambiguity:** A uniform penalty treats all non-target classes equally, ignoring that some classes are inherently confusable within a single view. For example, in the 'Intra View' panel, a 'Cylinder' and 'Cube' might project to similar square-like representations. Adopting a uniform regularization would depress the model's ability to express such inherent uncertainty.
- (2) **Inter-View Quality Gaps:** A single global weight is applied across all views, disregarding significant differences in view reliability. As illustrated in the 'Inter View' panel, the same object can yield both a 'High Quality' view and a 'Low Quality' (e.g., noisy or corrupted) view. The uniform design fails to account for this, allowing the 'Low Quality' view to degrade overall performance.

To overcome these issues, we propose a view-specific adaptive regularization strategy, which tailors the KL penalty to each view's unique biases along two dimensions:

- **Relax KL by class ambiguity.** We generate class similarity matrices from single-view classifier weights. Then, we use this similarity to **softly relax** the KL penalty on non-target classes that are representation-wise similar to the ground-truth, while keeping strong regularization on clearly separable classes.
- **Scale KL by view confidence.** We leverage evidential uncertainty as a proxy for view quality to dynamically scale the KL strength. This imposes stronger constraints on highly uncertain (low-quality) views to suppress noise, while relaxing penalties on confident (high-quality) views.

## 2 Method

### 2.1 Preliminaries

*Single View Evidential Modeling.* Given a multi-view dataset  $\mathcal{D} = \{\{x_i^{(v)}\}_{v=1}^V, y_i\}_{i=1}^N$  with  $N$  samples,  $V$  views, and  $K$  classes, where  $x_i^{(v)}$  and  $y_i$  denote the  $v$ -th view and label of the  $i$ -th sample, respectively. TMC use a separate neural network to generate non-negative evidence  $\mathbf{e}_i^{(v)}$  for  $v$ -view representation for sample  $i$ , which formulates a view-specific Dirichlet distribution with parameters  $\alpha_i^{(v)} = \mathbf{e}_i^{(v)} + 1$ .

*Dempster-Shafer Fusion.* In Subjective Logic [9], view representation induces a belief mass  $b_{ik}^{(v)} = (\alpha_{ik}^{(v)} - 1)/S_i^{(v)}$  that quantifies the confidence in the view's prediction for class  $k$  and an overall uncertainty  $u_i^{(v)} = K/S_i^{(v)}$ , where  $S_i^{(v)} = \sum_k \alpha_{ik}^{(v)}$  denote the Dirichlet strength. The belief masses and uncertainty generate an opinion over the class space  $M_i^{(v)} = [b_{i1}^{(v)}, \dots, b_{iK}^{(v)}, u_i^{(v)}]$ .

To fuse all view opinions, TMC methods typically employ the Dempster-Shafer (DS) combination rule [7]. We follow this rule and obtain a fused opinion  $M_i^F = M_i^{(1)} \oplus \dots \oplus M_i^{(V)}$ , corresponding to a fused Dirichlet vector  $\alpha_i^F$ . In the two-view case with  $M_i^1 = [b_{i1}^1, \dots, b_{iK}^1, u_i^1]$  and  $M_i^2 = [b_{i1}^2, \dots, b_{iK}^2, u_i^2]$ , the fusion is determined by the conflict mass  $C_i = \sum_{p \neq q} b_{ip}^1 b_{iq}^2$  and

$$b_{ik} = \frac{b_{ik}^1 b_{ik}^2 + b_{ik}^1 u_i^2 + b_{ik}^2 u_i^1}{1 - C_i}, \quad u_i = \frac{u_i^1 u_i^2}{1 - C_i}. \quad (1)$$

We then map the fused opinion  $M_i^F = [b_{i1}, \dots, b_{iK}, u_i]$  back to Dirichlet parameters by  $S_i^F = K/u_i$  and  $\alpha_{ik}^F = b_{ik} S_i^F + 1$ , yielding  $\alpha_i^F = (\alpha_{i1}^F, \dots, \alpha_{iK}^F)$  used in our subsequent loss design.

*Vanilla TMC Loss.* The overall  $\mathcal{L}_{\text{TMC}}$  objective [7] combines the loss on the fused opinion with the losses from all individual views:

$$\mathcal{L}_{\text{TMC}} = \frac{1}{N} \sum_{i=1}^N \left( \mathcal{L}_{\text{vanilla},i}(\alpha_i^F) + \sum_{v=1}^V \mathcal{L}_{\text{vanilla},i}(\alpha_i^{(v)}) \right). \quad (2)$$

The per-sample loss  $\mathcal{L}_{\text{vanilla},i}$  for any generic parameter vector  $\alpha_i$  (e.g.,  $\alpha_i^F$  or  $\alpha_i^{(v)}$ ) and label  $y_i$  consists of two components:

$$\mathcal{L}_{\text{vanilla},i}(\alpha_i) = \mathcal{L}_{\text{cls},i}(\alpha_i) + \lambda_t \mathcal{L}_{\text{KL},i}(\alpha_i), \quad (3)$$

where  $\lambda_t$  is an annealing coefficient that ramps from 0 to 1 over a predefined number of epochs and remains 1 thereafter.

The first component is the evidential classification loss, which serves as the primary data-fitting objective to maximize evidence for the true class:

$$\mathcal{L}_{\text{cls},i}(\alpha_i) = \sum_{k=1}^K y_{ik} (\log S_i - \log \alpha_{ik}). \quad (4)$$

The second component is the KL regularizer, whose purpose is to penalize and suppress evidence assigned to incorrect (non-target) classes:

$$\mathcal{L}_{\text{KL},i}(\alpha_i) = \text{KL}(\text{Dir}(\tilde{\alpha}_i) \parallel \text{Dir}(1)), \quad (5)$$

where  $\tilde{\alpha}_i = y_i + (1 - y_i) \odot \alpha_i$  directs the penalty only to non-target classes. The core limitation we address is that this  $\mathcal{L}_{\text{KL},i}$  (Eq. 5) is *uniform*, ignoring the view-specific biases we aim to model.

### 2.2 Intra-View Similarity-Relaxation

To make the KL term sensitive to each view's unique class structure, we use a separate relaxation matrix for every individual view.

*Similarity relaxation matrices.* To capture view-specific class confusion, we derive semantic correlations from the classifier's weights. Treating these weights as class prototypes [1, 13], we construct a similarity matrix reflecting the inherent representational overlap. For view  $v$ , let

$$W_v \in \mathbb{R}^{K \times D_v} \quad (6)$$

denote the classifier weight matrix, whose  $i$ -th row  $w_{v,i}$  corresponds to class  $i$ . We interpret  $w_{v,i}$  as a class prototype and compute cosine similarities between prototypes. We first  $\ell_2$ -normalize each row and stack them into

$$\hat{w}_{v,i} = \frac{w_{v,i}}{\|w_{v,i}\|_2}, \quad \hat{W}_v = \begin{bmatrix} \hat{w}_{v,1}^\top \\ \vdots \\ \hat{w}_{v,K}^\top \end{bmatrix}. \quad (7)$$

The resulting similarity matrix is

$$\text{Sim}_v = \hat{W}_v \hat{W}_v^\top \in \mathbb{R}^{K \times K}, \quad (8)$$

whose entries are cosine similarities in  $[-1, 1]$ . They capture how easily two classes can be confused in the representation space of view  $v$ , rather than a fixed global semantic prior. We map it to  $[0, 1]$  via an affine transform:

$$R_v = \frac{\text{Sim}_v + 1}{2}, \quad (9)$$

and use  $R_v$  as the semantic relaxation matrix for view  $v$ .

*Class-wise KL relaxation.* For a given view  $v$  and sample  $i$  with label  $y_i$ , we retrieve the row corresponding to  $y_i$  from the relaxation matrix  $R_v$ :

$$\mathbf{r}_i^{(v)} = R_{v,y_i,:} \in [0, 1]^K, \quad (10)$$

and define a class-wise mask vector

$$\mathbf{m}_i^{(v)} = 1 - \mathbf{r}_i^{(v)} \in [0, 1]^K, \quad (11)$$

By construction  $r_{i,y_i}^{(v)} = 1$  and thus  $m_{i,y_i}^{(v)} = 0$ , so the true class is not regularized by KL, as in the vanilla design.

The *relaxed* Dirichlet parameter vector is then

$$\boldsymbol{\alpha}_i^{(v)'} = (\boldsymbol{\alpha}_i^{(v)} - 1) \odot \mathbf{m}_i^{(v)} + 1 = \mathbf{e}_i^{(v)} \odot \mathbf{m}_i^{(v)} + 1, \quad (12)$$

For non-target classes, large entries in  $\mathbf{r}_i^{(v)}$  (closer to the ground-truth class in the view- $v$  feature space) yield small entries in  $\mathbf{m}_i^{(v)}$  and thus a weak KL penalty, whereas unrelated classes (with  $r_{ik}^{(v)} \approx 0$ ) keep  $m_{ik}^{(v)} \approx 1$  and receive strong regularization. Replacing the original non-target parameters in the standard KL with  $\boldsymbol{\alpha}_i^{(v)'}$  yields an intra-view *similarity-relaxed* KL:

$$\mathcal{L}_{\text{Relaxed-KL},i}^{(v)} = \text{KL}(\text{Dir}(\boldsymbol{\alpha}_i^{(v)'}) \parallel \text{Dir}(\mathbf{1})). \quad (13)$$

This similarity-relaxed KL suppresses evidence on clearly unrelated classes while preserving moderate evidence for representation-wise ambiguous ones.

### 2.3 Inter-View Uncertainty Scaling

While the similarity-relaxed KL adapts regularization along the *class* dimension, the multi-view setting naturally provides a second axis: *view-specific uncertainty*. To capture the *relative* reliability among views, we normalize the uncertainty masses  $u_i^{(v)}$  across views of the same sample. We first compute the average uncertainty

$$\bar{u}_i = \frac{1}{V} \sum_{v=1}^V u_i^{(v)}, \quad (14)$$

and define an uncertainty scaling factor for each view

$$\rho_i^{(v)} = \frac{u_i^{(v)}}{\bar{u}_i}, \quad (15)$$

so that the average scaling across views satisfies  $\frac{1}{V} \sum_{v=1}^V \rho_i^{(v)} = 1$ . Views that are more uncertain than the average ( $\rho_i^{(v)} > 1$ ) are assigned a stronger KL penalty, while relatively confident views ( $\rho_i^{(v)} < 1$ ) are regularized more softly.

Given the similarity-relaxed KL  $\mathcal{L}_{\text{KL-relax},i}^{(v)}$  defined in the previous subsection, the resulting view-specific KL for view  $v$  and sample  $i$  is

$$\mathcal{L}_{\text{view-specific-KL},i}^{(v)} = \rho_i^{(v)} \mathcal{L}_{\text{Relaxed-KL},i}^{(v)}. \quad (16)$$

### 2.4 Overall Objective

Based on the vanilla fused loss  $\mathcal{L}_{\text{vanilla}}^F(i)$  and our new view-specific loss

$$\mathcal{L}_{\text{view-specific},i}^{(v)} = \mathcal{L}_{\text{cls},i}^{(v)} + \mathcal{L}_{\text{view-specific-KL},i}^{(v)}, \quad (17)$$

our final objective averages over the training set:

$$\mathcal{L}_{\text{ours}} = \frac{1}{N} \sum_{i=1}^N \left( \mathcal{L}_{\text{vanilla},i}^F + \sum_{v=1}^V \mathcal{L}_{\text{view-specific},i}^{(v)} \right). \quad (18)$$

This objective keeps the standard evidential supervision on the fused opinion while introducing view-specific, similarity-relaxed and uncertainty-scaled KL regularization at the view level.

## 3 Experiments

### 3.1 Experimental Setup

**Datasets:** We evaluate our method on several real-world multi-view datasets, including CUB [15], PIE<sup>1</sup>, Scene15 [5], Caltech101 [4], Handwritten [14], and Animal [10].

**Baselines:** We compare our method (Ours-AR) against traditional multi-view and TMC models, including DCCAE [16], CPM-Nets [20], DUA-Nets [6], TMC [7], ETMC [8], SMDC [12], ECML [17], and RMVC [19].

**Implementation Details.** For each view, we append a fully connected layer and apply a softplus activation to map the evidence into  $[0, \infty)$ . The network is trained using the Adam optimizer with a weight decay of  $1 \times 10^{-5}$ . For all datasets, 20% of the samples are used as test sets. We report the mean results across five runs.

### 3.2 Performance Comparison

As shown in Table 1, our method achieves the best performance on five datasets and second-best on CUB. In particular, we achieve 1.75% improvement over the strongest competitor on PIE, while remaining highly competitive on CUB with only 0.55% behind ECML. These results demonstrate that explicitly modeling class-wise semantic relations and view-wise quality does not trade off standard recognition performance; instead, it consistently strengthens the overall multi-view classifier.

**Table 1: Comparison with multi-view classification methods on accuracy (%). Best in bold, second-best underlined.**

| Method   | Handwritten  | CUB          | Caltech101   | PIE          | Scene15      | Animal       |
|----------|--------------|--------------|--------------|--------------|--------------|--------------|
| DCCAE    | 97.27        | 85.44        | 77.33        | 82.43        | 55.76        | 86.30        |
| CPM-Nets | 95.74        | 89.73        | 83.70        | 81.60        | 68.32        | 84.12        |
| DUA-Nets | 97.52        | 71.65        | 84.27        | 81.76        | 65.01        | 78.65        |
| TMC      | 98.03        | 91.47        | <u>89.19</u> | 88.26        | 68.13        | 87.20        |
| ETMC     | 98.50        | 92.34        | 88.66        | 91.60        | 72.86        | <u>88.21</u> |
| SMDC     | 97.05        | 88.50        | 87.20        | 89.11        | 70.42        | 87.59        |
| ECML     | <u>98.72</u> | <b>94.27</b> | 88.84        | <u>92.32</u> | <u>73.84</u> | 87.25        |
| RMVC     | 98.51        | 93.18        | 88.73        | 91.18        | 73.05        | 87.67        |
| Ours     | <b>99.05</b> | <u>93.72</u> | <b>90.78</b> | <b>94.07</b> | <b>75.35</b> | <b>89.43</b> |

### 3.3 Ablation Study

We further conduct an ablation study on Scene15 to analyze the contribution of the two components in our adaptive regularization strategy. As shown in Table 2, the baseline objective without either component achieves 68.13% accuracy. Enabling only the semantic relaxation term improves the performance to 73.32%, while enabling only the uncertainty-weighted term yields 71.64%. When both components are activated simultaneously, the accuracy further increases to 75.35%, indicating that they address complementary aspects of view-specific bias and work best when combined.

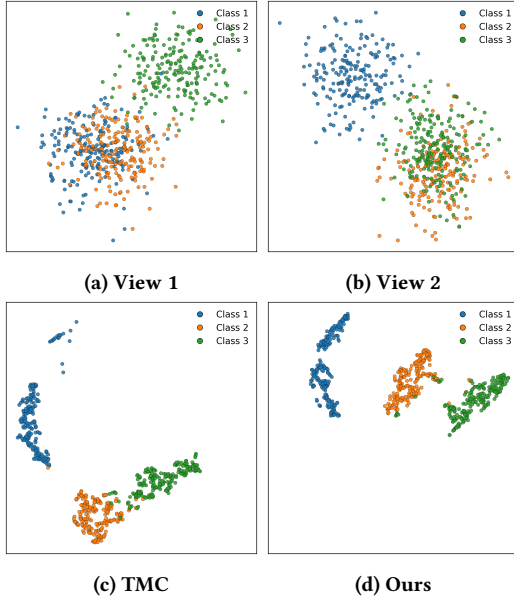
<sup>1</sup><http://www.cs.cmu.edu/afs/cs/project/PIE/MultiPie/Multi-Pie/Home.html>

**Table 2: Ablation study on Scene15 Dataset.**

| $\mathcal{L}_{intra}$ (Relaxation) | $\mathcal{L}_{inter}$ (Quality) | Scene15      |
|------------------------------------|---------------------------------|--------------|
| ×                                  | ×                               | 68.13        |
| ✓                                  | ×                               | 73.32        |
| ×                                  | ✓                               | 71.64        |
| ✓                                  | ✓                               | <b>75.35</b> |

### 3.4 Analysis on Synthetic Data

We first construct a 2D toy dataset with three classes and two views to highlight view-specific ambiguity and complementarity. In view 1, classes 1 and 2 are strongly overlapped while class 3 is more separated; in view 2, the ambiguity shifts to classes 2 and 3, so each single view is highly confusing. On this toy example, TMC achieves **95.84%** accuracy after fusion, whereas our adaptive regularization improves the fused accuracy to **97.50%**. For visualization, we extract the model’s evidence representations and apply t-SNE to project them into 2D. As shown in Figure 2, our method produces more compact and better separated clusters, indicating that it can down-weight ambiguous views while preserving complementary information.

**Figure 2: Visualization on the synthetic toy dataset.**

## 4 Conclusion

In this paper, we proposed an adaptive KL regularization scheme for trusted multi-view classification. Instead of a uniform KL objective, our method introduces a view-specific similarity-relaxed KL that softens penalties on semantically related non-target classes, and an uncertainty-scaled KL that scales regularization by view reliability. Experiments show consistent gains over strong baselines, and ablations verify that the two components are complementary. Additional toy visualizations provide qualitative support. Future work

will extend this adaptive regularization to scenarios with missing views and more challenging uncertainty settings.

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